Sheet 55 Thu. 25 March 2027 Critical reading of media formulations

1. Warm-up · Flash [PRO] A 10 % increase on 40: what final value?

Ans. $40 \times 1{,}10 = 44$.

2. Core · Critical thinking [LEC] An article states: "60 % of accidents involve sober drivers". Can we conclude that driving sober is dangerous?

Ans. No: the reference population is missing. If the vast majority of drivers are sober, it is normal for them to be the majority among those involved in accidents. Conditional frequencies should be compared.

3. Core + · True/false [LEC] True or false? Justify. "Two out of three smokers are men" means that two out of three men smoke.

Ans. False. This confuses $_{\text{smoker}}(\text{man})$ and $_{\text{man}}(\text{smoker})$: the reference populations differ.

4. Back · True/false [STA] True or false? Justify. The median of a series is always equal to its mean.

Ans. False. Counterexample: 1, 2, 9: mean 4, median 2.

5. Running thread · Reading code [ALG] What is the value of `s` at the end?

```
s = 0
for i in range(1, 5):
    s = s + i
```

Ans. $s = 1 + 2 + 3 + 4 = 10$.

Pitfall: range(1, 5) stops at 4 ; 5 is excluded.

Coordinate geometry: vectors and lines review

Sheet 56 Tue. 30 March 2027 Vectors and collinearity: review

1. Warm-up · Flash [GEO] $\vec{u}\begin{pmatrix}3\\-2\end{pmatrix}$ & $\vec{v}\begin{pmatrix}-1\\5\end{pmatrix}$. Calculate $\vec{u}+\vec{v}$ and $2\vec{u}$.

Ans. $\begin{pmatrix}2\\3\end{pmatrix}$; $\begin{pmatrix}6\\-4\end{pmatrix}$.

2. Core · Method [GEO] Are points $A(-1;2)$, $B(1;3)$ and $C(5;5)$ collinear?

Ans. $\overrightarrow{AB}\begin{pmatrix}2\\1\end{pmatrix}$, $\overrightarrow{AC}\begin{pmatrix}6\\3\end{pmatrix}$: $2\times 3 - 1\times 6 = 0$: collinear.

3. Core + · Method [GEO] $A(-1;2)$, $B(3;1)$, $C(4;4)$. Determine D such that $ABCD$ is a parallelogram.

Ans. $\overrightarrow{AB} = \overrightarrow{DC}$: $D = A + \overrightarrow{BC} = (-1+1;2+3) = (0;5)$.

4. Back · Flash [PRO] In a class of 24 pupils, 18 are half-boarders. What proportion is this as a percentage?

Ans. $\dfrac{18}{24} = 0{,}75 = 75\,\%$.

5. Running thread · Prove [LIT] Show that $(n+1)^2 - (n-1)^2 = 4n$ for every integer n. What can we conclude about the difference of the squares of two consecutive odd numbers (n even)?

Ans. $(n^2+2n+1) - (n^2-2n+1) = 4n$. For $n = 2k$ even, the diff. of squares of two consecutive odd numbers is 8 k : a multiple of 8.

Both integers n -1 and $n+1$ are odd when n is even.

Sheet 57 Thu. 1 st Apr. 2027 Determinant and equations of lines: review

1. Warm-up · Flash [GEO] Slope of the line $3x + 2y - 6 = 0$?

Ans. $y = -\dfrac{3}{2}x + 3$: slope $-\dfrac{3}{2}$.

2. Core · Method [GEO] Equation of the line (AB) with $A(1;2)$ and $B(4;8)$.

Ans. Slope $\dfrac{8-2}{4-1} = 2$; $y = 2x + p$, $2 = 2 + p$: $y = 2x$; or $2x - y = 0$.

3. Core + · Method [GEO] Are the lines $x - y + 1 = 0$ and $4x - 2y + 5 = 0$ parallel, coincident or intersecting?

Ans. Direction vectors $\begin{pmatrix}1\\2\end{pmatrix}$ and $\begin{pmatrix}2\\4\end{pmatrix}$ are collinear: parallel; $(0;1)$ satisfies the first but $4 \times 0 - 2 + 5 = 3 \neq 0$: strictly parallel.

4. Back · Flash [PUI] Calculate $10^{-2} \times 5 \times 10^{4}$.

Ans. $5 \times 10^{2} = 500$.

5. Running thread · Method [EQ] Solve in $\mathbb{R}$: $7 - 2(x-1) = 3x - 6$.

Ans. $x = 3$

Pitfall: $-2(x-1) = -2x - 2$ (sign error).

Sheet 58 Mon. 5 Apr. 2027 Plane geometry problems: choosing a method

1. Warm-up · Flash [GEO] Distance AB for $A(0;0)$ and $B(5;12)$?

Ans. $\sqrt{25+144} = 13$.

2. Core · Method [DEM] $A(0;0)$, $B(4;2)$, $C(-2;4)$. Type of triangle ABC ?

Ans. $AB^2 = 20$, $AC^2 = 20$, $BC^2 = 36 + 4 = 40 = AB^2 + AC^2$: isosceles right-angled at A.

3. Core + · Method [DEM] Give two methods to prove that $ABCD$ is a parallelogram using coordinates.

Ans. (i) $\overrightarrow{AB} = \overrightarrow{DC}$; (ii) the diagonals $[AC]$ and $[BD]$ have the same midpoint.

4. Back · Reading [LEC] In a class of 30 pupils, a bar chart shows x pupils with mark 12 and $2x$ pupils with mark 14; the other marks account for 18 pupils. Find x.

Ans. $x + 2x + 18 = 30$, hence $3x = 12$ and $x = 4$.

5. Running thread · Flash [SOL] A pyramid has a square base of side 6 cm and height 5 cm. Calculate its volume.

Ans. $V = \dfrac{1}{3} \times 6^2 \times 5 = 60 \text{ cm}^3$.

Sheet 59 Tue. 6 Apr. 2027 Situations from other disciplines

1. Warm-up · Flash [MOD] A moving point has position $x(t) = 3t + 2$ (in m, t in s). Where is it at $t = 4$?

Ans. $x(4) = 14$ m.

2. Core · Method [MOD] A second moving point follows $x_2(t) = 5t - 4$. At what time and where do the moving points meet?

Ans. $3t + 2 = 5t - 4 \Leftrightarrow t = 3$ s; position $x = 11$ m.

3. Core + · Flash [GEO] A velocity vector has coordinates $\begin{pmatrix}3\\4\end{pmatrix}$ (in m/s). What is its magnitude?

Ans. $\sqrt{3^2 + 4^2} = 5$ m/s.

4. Review · Flash [CAL] Calculate $\dfrac{1}{2} - \dfrac{3}{5}$.

Ans. $\dfrac{5}{10} - \dfrac{6}{10} = -\dfrac{1}{10}$.

5. Running thread · Write up [DEM] ABC is isosceles at A and H is the foot of the altitude from A. What can we say about H ?

Ans. H is the midpoint of $[BC]$ (the altitude from the apex is also the perpendicular bisector of the base).

Sheet 60 Thu. 8 Apr. 2027 Communicating a proof: implication, converse, equivalence

1. Warm-up · Flash [DEM] State the converse of: "if a quadrilateral is a rhombus, then its diagonals are perpendicular".

Ans. "If a quadrilateral has perpendicular diagonals, then it is a rhombus."

2. Core · True/false [DEM] True or false? Justify. The previous converse is true.

Ans. False. Counterexample: a kite (not a rhombus) has perpendicular diagonals.